
LEARNING TO WALK EFFICIENTLY WITH TD3

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ABSTRACT

This paper uses Twin Delayed Deep Deterministic Policy Gradient (TD3) to train a bipedal walker agent in both the standard and hardcore environments. Key modifications include LayerNorm, small-weight output initialisation, reward clipping, and exploration noise decay for the standard environment, with additional random action warmup and gradient clipping for the hardcore environment. In the standard environment, the agent begins achieving positive rewards around episode 166 and reaches a peak score of 238.2 at episode 903. In the hardcore environment, the agent first scores positively around episode 318 and reaches a peak of 227.8 at episode 1898.

1 METHODOLOGY

TD3 is used as the base algorithm because the Walker task has a continuous action space and TD3 is a widely used off-policy method for such problems. TD3 uses two critic networks and takes the smaller Q-value to reduce overestimation bias. The actor is updated less frequently than the critics to improve training stability. In addition, small noise is added to target actions (target policy smoothing) to prevent the policy from exploiting sharp errors in the learned Q-function. The target value is computed as [1]:

$$y = r + \gamma \min_{i=1,2} Q_{\theta'_i}(s', \pi_{\phi'}(s') + \epsilon), \quad \epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c) \quad (1)$$

where $\gamma = 0.99$ is the discount factor, $\sigma = 0.2$ is the target noise scale, and $c = 0.5$ is the noise clip range.

Both critics are trained jointly by minimising [1]:

$$L_{\text{critic}} = \frac{1}{N} \sum_{i=1}^N \left[(Q_{\theta_1}(s_i, a_i) - y_i)^2 + (Q_{\theta_2}(s_i, a_i) - y_i)^2 \right] \quad (2)$$

where y_i is the target value from Equation (1).

The actor network has two hidden layers of 400 and 300 units with ReLU activations and a tanh output layer. The critic consists of two Q-networks, each taking the concatenation of state and action as input through the same 400-300 structure. Exploration noise is added to actions during training.

1.1 DESIGN CHOICES COMMON TO BOTH ENVIRONMENTS

Reward clipping. A minimum reward of -10 is applied in code. Without clipping, the agent receives -100 when it falls, which creates large target values that destabilise the critic.

Exploration noise scheduling. Exploration noise is adjusted during training to balance exploration and stability. Higher noise is used in earlier stages to encourage exploration, while lower noise is used later to produce smoother and more stable actions. For standard environment, Exploration noise is reduced from 0.1 to 0.05 after episode 700. For hardcore environment, Exploration noise decays linearly from 0.1 to 0.01 between episodes 500 and 2000.

1.2 DESIGN CHOICES FOR THE STANDARD ENVIRONMENT

LayerNorm. LayerNorm [2] is applied after the first hidden layer in both the actor and critic. LayerNorm normalises the activations within a layer to have zero mean and unit

variance, which helps reduce the effect of differing feature scales and can make training more stable.

Small weight initialisation. Following [3], the output layers are initialised with small uniform weights ($\pm 3 \times 10^{-3}$). This ensures the actor produces near-zero actions initially, allowing the agent to explore gradually rather than making large movements that cause immediate falls.

Action scaling. Actions are scaled by 0.95 after episode 700. Scaling actions by 0.95 reduces the torque applied at each step, which lowers the torque penalty and results in higher scores.

1.3 DESIGN CHOICES FOR THE HARDCORE ENVIRONMENT

Removing LayerNorm and small weight initialisation. LayerNorm and small weight initialisation were removed as they did not improve performance in the hardcore environment. The harder terrain requires more aggressive initial exploration, which is hindered by near-zero initial actions. LayerNorm may also over-constrain the network when the agent needs to learn a wider range of behaviours to handle diverse obstacles.

Random action warmup. The first 25,000 steps use uniformly random actions instead of the actor network. In the hardcore environment, early episodes are very short as the agent falls quickly, filling the replay buffer with similar failure transitions. The warmup ensures the buffer contains diverse experiences before learning begins.

Gradual noise decay. Instead of a single noise reduction, exploration noise decays linearly from 0.1 to 0.01 between episodes 500 and 2000. The hardcore environment contains many different obstacle types, so the agent needs longer exploration to encounter and learn from all of them.

Gradient clipping. The critic gradient is clipped to a maximum norm of 1. The hardcore environment produces more varied rewards due to obstacles, which can cause large gradient updates that destabilise training.

1.4 HYPERPARAMETERS

Table 1 summarises the hyperparameters for both environments.

Hyperparameter	Standard	Hardcore
LayerNorm	Yes	No
Small weight init	Yes	No
Learning rate	3×10^{-4}	3×10^{-4}
Batch size	256	256
Discount γ	0.99	0.99
Soft update τ	0.005	0.005
Policy delay	2	2
Replay buffer	10^6	10^6
Exploration noise	$0.1 \rightarrow 0.05$	$0.1 \rightarrow 0.01$
Random warmup	No	25,000 steps
Gradient clipping	No	norm = 1
Reward clipping	min = -10	min = -10

Table 1: Hyperparameters for standard and hardcore environments.

2 CONVERGENCE RESULTS

2.1 STANDARD ENVIRONMENT

Figure 1 shows the training curve for the standard environment over 1000 episodes. The agent first achieves a positive reward at episode 166. By episode 283, the rolling-50 mean

exceeds 200, indicating convergence. After episode 700, exploration noise is reduced from 0.1 to 0.05, which makes the policy more stable with only 3 episodes falling below zero in the final 300 episodes. The peak score of 238.2 is reached at episode 903.

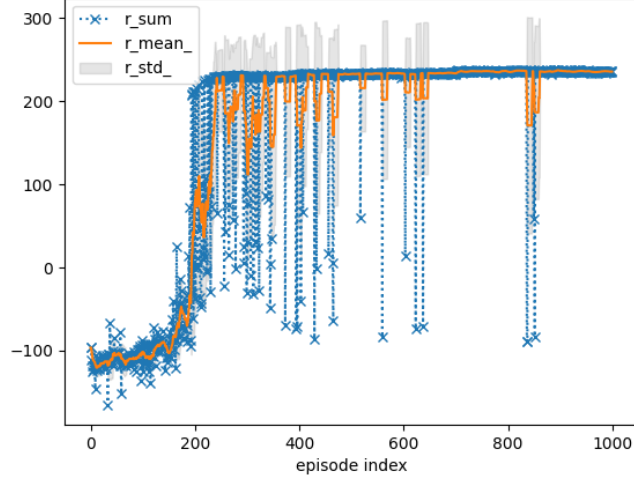


Figure 1: Training curve for the standard environment over 1000 episodes.

2.2 HARDCORE ENVIRONMENT

Figure 2 shows the training curve for the hardcore environment over 2000 episodes. Due to the 25,000-step random warmup, learning begins around episode 40. The agent first achieves a positive reward at episode 318. By episode 1134, the rolling-50 mean exceeds 200, indicating convergence. In the final 500 episodes, 18 still result in negative scores, likely caused by difficult terrain features such as stairs and pitfalls. The peak score of 227.8 is achieved at episode 1898.

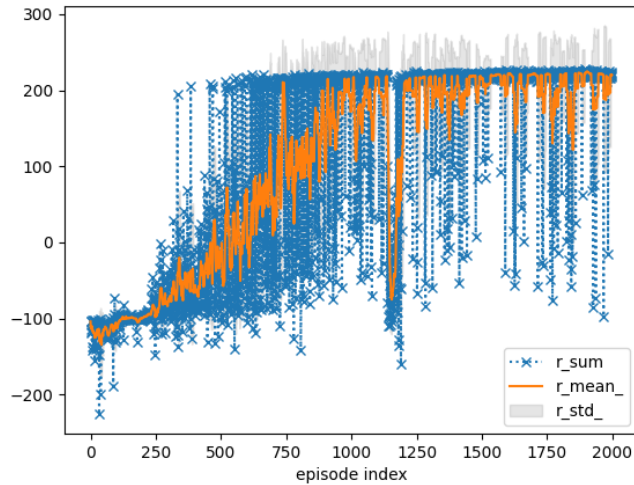


Figure 2: Training curve for the hardcore environment over 2000 episodes.

3 LIMITATIONS AND FUTURE WORK

In the standard environment, the agent reaches a peak score of 238.2 but does not reach the environment ceiling of approximately 250. The policy converges to a conservative gait that prioritises stability over speed. Action scaling by 0.95 reduces the torque penalty but also limits the maximum speed the agent can achieve.

In the hardcore environment, the agent still fails on some difficult terrain features, with 18 negative-score episodes in the final 500. This indicates the policy is not fully robust to all obstacle types such as stairs and pitfalls.

For future work, prioritised experience replay [4] could be used to replace uniform sampling from the replay buffer. This method samples transitions based on TD-error magnitude, so that transitions with larger prediction errors are replayed more frequently. This may help the agent learn more effectively from rare but important experiences, such as encountering obstacles in the hardcore environment..

USE OF GENERATIVE AI

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I used ChatGPT to assist with analysing the training log files, such as computing rolling averages, identifying convergence points (e.g. the episode at which the rolling-50 mean exceeds 200), and counting failure episodes in the final training stages. I also used it to suggest potential improvements for the future work section, such as prioritised experience replay.

REFERENCES

- [1] Scott Fujimoto, Herke Hoof, and David Meger. “Addressing function approximation error in actor-critic methods”. In: *International Conference on Machine Learning*. PMLR. 2018, pp. 1587–1596.
- [2] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. “Layer normalization”. In: *arXiv preprint arXiv:1607.06450* (2016).
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- [4] Tom Schaul et al. “Prioritized experience replay”. In: *arXiv preprint arXiv:1511.05952* (2015).